

RESEARCH ARTICLE

Overall Material Flow Scheme Design of Controlled Ecological Life Support System for Lunar Base

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Recently, the Moon has once again become the focus of the aerospace field. The lunar base is likely to be the next long-term extraterrestrial base for humans after the near-Earth orbit space station. However, the cost of lunar replenishment will substantially increase, and the time required for replenishment will become longer. It is necessary to establish a living system with higher degree of material closure and stronger self-sustainability to provide life support service for residents. The controlled ecological life support system (CELSS) is a feasible way to achieve the circular supply of oxygen, water, and food through biological regeneration. However, there is a practical issue of how to gradually establish a CELSS on the Moon and how to match the bio-regenerative system with the physicochemical system established in the early stage, to reduce the cost of the construction of the lunar life support system. In this paper, based on fully inheriting the physicochemical regeneration technologies of the present near-Earth orbit space station, biological components such as plants, microalgae, and microorganisms were gradually introduced in 4 stages to establish an upgraded CELSS with a total material closure degree of 98%. In addition, the material flux models of C, H, and O elements at different stages of the lunar base were designed and calculated. The results of this paper provide a reference for the construction of future lunar base life support systems in terms of improving the engineering feasibility of CELSS.

Introduction

With the development of manned space technology in recent years, people have once again shifted their attention to further extraterrestrial spaces such as the Moon and Mars. NASA has proposed the Artemis project, which planned to establish suitable workplaces for long-term human habitation in lunar orbit or on the lunar surface [1]. The European Space Agency also proposed the construction concept of an open lunar base—Moon Village [2], and participated in the Artemis project. Based on the completion, design, and construction of the Tiangong space station in near-Earth orbit [3], China has also proposed a corresponding manned lunar landing plan and will jointly build an International Lunar Research Station with Russia [4].

The life support system is the fundamental premise of manned space missions, and the near-Earth orbit space station (International Space Station and China Space Station) achieved the regeneration supply of oxygen and water resources through physicochemical technologies, which greatly reduced the ground replenishment [3,5]. As human spaceflight develops in the direction of deep space exploration, such as the Moon, the increased difficulty and reduced timeliness of material supply have brought new challenges to the life support system. There is a need to improve the material closure of the system, reduce dependence

on terrestrial resupply, and minimize the impact on the Moon's native environment. The controlled ecological life support system (CELSS) can achieve the simultaneous regeneration of oxygen, water, and food; improve the material closure of the system; and provide psychological support by creating a green and fresh natural environment for human beings [6,7]. In addition, biological components have the ability to self-repair, which can further enhance the system's self-sustainability. Therefore, the CELSS is an effective way to accomplish life support for lunar base missions [8–11].

For the functional design and material flow regulation of extraterrestrial CELSS, previous researchers have carried out a few designs in conjunction with the construction and operation of ground-based experimental systems. Vermeulen et al. [12] developed stoichiometric equations for various processes including anaerobic fermentation of waste, photosynthetic heterotrophic synthesis, urea hydrolysis and nitrification, plant and microalgae growth, and respiratory metabolism of personnel based on the system-building model of MELiSSA (Micro-Ecological Life Support System Alternative) [13], and proposed a material flow model of the MELiSSA system with 100% material closure for a 6-person crew. Based on the research on the vegetable production system named EDEN in Antarctica, Maiwald et al. [14,15] proposed the construction scheme of the ground-based test system

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of the lunar greenhouse and formulated the architecture of the test system especially focusing on the design of the plant system. Hinterman et al. [16] designed the Martian Garden, a scalable, closed-loop ecosystem for 4 people, and provided structural and systems engineering design. Hu et al. [17] proposed to construct a bio-regenerative life support system based on 15 species of plants and silkworms, as well as functional components such as biological treatment of waste. Quantitative calculations of the material flow of the system were also performed. Previous studies on the design of system architecture and material flow balance of CELSS mainly focused on the modeling of biological components and the bio-regeneration system, thus ignoring the following 2 practical issues. Firstly, the construction of the lunar base will be completed gradually. With the step-by-step implementation of manned lunar exploration, the life support system will gradually transition from the non-regenerative type of short-term residence in the early stage to the physicochemical regenerative type and even the bio-regenerative system of long-term multi-person residence. The material flow design in the construction process needs to be considered. Secondly, bio-regenerative components have their own inherent growth rhythm. Biological components do not have sufficient material production capacity at the initial stage of growth to fulfill their life-supporting functions independently. It is necessary to establish a closed environment suitable for the survival of human and plant by means of physiochemical regeneration. The 4-subject 180-day CELSS integration test carried out in 2016 also verified this process [18]. Therefore, we believe that the construction of the CELSS needs to be carried out from the basic environment established through physiochemical regenerative system, and the collaborative matching between physiochemical regenerative components and bio-regenerative components should be considered in the construction process.

In this paper, we introduce biological components of different scales and types to design the matching CELSS for different stages of lunar exploration missions, explore the construction scheme of the CELSS in which bio-regeneration gradually replaces physiochemical regeneration, and complete the material flow scheme design of the CELSS at different stages and scales, to provide a reference for subsequent engineering design and system experimental design.

Methods

Basic concepts and rationale of CELSS

In this paper, we put forward the concept of CELSS from the perspective of the engineering application of the environmental control and life support system for manned space missions. That is, based on the basic principles of ecology of “producer–consumer–decomposer”, it constructs an artificial ecosystem of “human–(animal) plant–microbe–environment” in a closed environment, which is characterized by biological regeneration and organically integrates non-regenerative and physicochemical regeneration technologies. The system efficiently recycles life-supporting materials such as food, water, and oxygen through biological processes such as plant growth and microbial decomposition, supplemented by certain physicochemical means. The application of CELSS could substantially reduce logistics supplies for long-term manned space missions such as deep space exploration and extraterrestrial residence.

The characteristics of a CELSS mainly include the following: (a) In the initial phase of system construction, when the bio-regeneration capacity is limited, the fundamental environment

is established through non-regeneration and physicochemical regeneration technologies, while simultaneously achieving material regeneration. In the subsequent phase of system construction, the bio-regeneration system will assume a dominant role in material circulation, while the non-regeneration and physicochemical regeneration system will serve as a backup for material assurance; (b) Through biological processes such as plants, microalgae, and microorganisms, it is possible to achieve the regeneration and continuous supply of all necessary life support substances such as oxygen, water, and food. At the same time, it maximizes the utilization of material resources within the system and in situ resources from the extraterrestrial space, reducing its dependence on ground replenishment, thereby obviously enhancing its material closure and self-sustainability; (c) Due to the mutual support and redundancy of non-regeneration, physicochemical regeneration, and bio-regeneration systems in CELSS, as well as the good self-repair capability of biological components, the CELSS has high reliability and strong adaptability. It can address life support challenges in complex extraterrestrial environments and has marked potential for further development.

Overall design approach and subsystems composition

From the perspective of the development and application of life support technology in the International Space Station and China's space station, the life support system of early lunar base (medium-to long-term residency of 20 weeks to 20 months) should also be based on physicochemical regeneration technology, with a few bio-regeneration technologies [3,19], and the role of bio-regeneration will gradually increase. It should be noted that this paper focuses on the scheme design and quantitative model of the material flow of the regenerative life support system, while basic environmental control functions such as air pressure regulation, temperature and humidity control, trace harmful gas purification, and particulate and microbial control are not included in the design and calculation scope of this paper. In addition, the technologies selected in the overall material flow scheme of this paper are all mature and applicable technologies based on existing development levels. Immature technologies that have not yet been verified are not included such as lunar soil utilization, lunar ice collection and utilization, animal protein production, artificial starch synthesis, etc. These technologies will be fully studied and verified before being embedded in the system scheme.

The functions of each regeneration subsystem involved in this paper are shown in Table 1, and a brief description of the selection, configuration, and integration of each subsystem is given.

Results

Overall application strategy of the CELSS in the lunar base

The overall application strategy of CELSS in the lunar base is designed in 4 stages as below. The overall design principles mainly include the following: (a) In order to enhance the comprehensive technological maturity of the system, we prioritize the selection of subsystem technologies that have been successfully applied or fully validated. The subsystem technologies selected in this paper are derived from 2 primary sources. The first is the physicochemical regeneration technology that has been applied in China's space station [3], and the other is the

Table 1. Function and configuration of the regeneration subsystems for CELSS. For detailed information on regenerative subsystems 1 to 5 below, please refer to the report of Liu et al. [3]. For detailed information on regenerative subsystems 6 to 10 below, please refer to the report of Tang et al. [20].

ID	Subsystem	Function	Selection and integration methods
1	CO ₂ removal subsystem (CO ₂ removal)	Removes or enriches CO ₂ in the atmosphere through adsorption or absorption, and reduces the concentration level of CO ₂ in the atmosphere.	It is used in the earlier lunar base and will be switched to a backup in the later stage, when CO ₂ is capable to be removed from the atmosphere through biological photosynthesis. Its size is determined by the personnel number of prophase lunar base.
2	Electrolysis oxygen generation subsystem (oxygen generation)	O ₂ is generated by H ₂ O electrolysis, assuming a 100% conversion efficiency.	It is used in the earlier lunar base and will be switched to a backup in the later stage, when O ₂ will be produced through biological photosynthesis. Its size is determined by the personnel number of prophase lunar base.
3	CO ₂ reduction subsystem (CO ₂ reduction)	Through chemical catalytic process, CO ₂ and H ₂ are reduced to CH ₄ and H ₂ O, and water is recovered. Assuming a 100% efficiency of material conversion.	It is used in combination with the CO ₂ removal and oxygen generation subsystem in the earlier lunar base, and will be switched to a backup in the later stage when CO ₂ reduction will be completed through biological photosynthesis. Its size is determined by the personnel number of prophase lunar base.
4	Urine processing subsystem	Recovers most of the majority water in the urine through distillation.	It is used in conjunction with the brine processing subsystem in the earlier lunar base, and later combined with the wastewater bioreactor subsystem to extend itself service life. As the main means of removing inorganic salts such as NaCl from urine, this subsystem will play a role in all stages of the lunar base CELSS. Its size is determined by the number of personnel and will expand as the population increases.
5	Water recovery subsystem	Completes the deep purification of water through physical and chemical methods such as catalytic oxidation, adsorption, and ion exchange to produce potable water.	It is used in early lunar base to purify the distilled condensate from the urine processing and brine processing subsystems to produce drinking water. In the later stage, it is combined with the wastewater bioreactor subsystem to produce drinking water and reduce the material consumption of the subsystem itself. It is essential for the production and supply of potable water in CELSS of all phases of the lunar base. Its size is determined by the number of personnel and will expand as the population increases.
6	Food-vegetable production subsystem (vegetable production)	Synthesizes organic biomass through plant photosynthesis to produce fresh vegetables, and simultaneously completes CO ₂ reduction and O ₂ generation.	Due to the relatively high production efficiency, it is applied earlier than the grain production subsystem. It provides a small number of fresh fruits and vegetables in the early stage, and provides all the required fresh fruits and vegetables in the later stage. Simultaneously, it completes equivalent CO ₂ removal and O ₂ generation. The size of its configuration is determined by the number of personnel and the closure of vegetables.

(Continued)

Table 1. (Continued)

ID	Subsystem	Function	Selection and integration methods
7	Food-grain production subsystem (grain production)	Synthesize organic biomass through plant photosynthesis to produce staple food, and simultaneously complete CO ₂ reduction and O ₂ generation.	There is no food production in the early lunar base, the grain production subsystem will be introduced and gradually expanded in the later stage. The gradually enhanced photosynthesis will replace the functions of the physicochemical regeneration subsystems including CO ₂ removal, oxygen generation, and CO ₂ reduction. The size of the configuration is determined by the number of personnel and food closure.
8	Solid waste dewatering/grinder subsystem (waste dewatering/grinder)	Solid waste such as inedible parts of plants is dried through internal circulation heating, and all the liquid water in the solid waste is recovered, and then the solid waste is crushed to the particle size required for further treatment.	For the CELSS system with low food closure, only the waste dewatering module is configured. When the food closure is increased, the grinder module is configured to complete the pretreatment before further transformation of solid waste. The configuration scale is determined by the output of inedible plants in the system.
9	Wastewater bioreactor subsystem (wastewater bioreactor)	Through microbial metabolism, wastewater bioreactor completes the pretreatment of wastewater such as urine, sanitary cleaning wastewater, and air condensate, removes organic pollutants from the wastewater, and completes nitrogen oxidation. Effluents meet the standards for plant cultivation or can enter the water recovery subsystem for further purification.	It is used in the mid-later stages' lunar base as a pretreatment and water circulation transfer station. Its purpose is to reduce material consumption and improve system efficiency and applicability. Its configuration scale is determined by the number of personnel and the scale of plant cultivation.
10	Solid waste bioreactor subsystem (waste bioreactor)	Through biological fermentation and conversion treatment, the biomass solid waste in the system (mainly inedible plant parts and crew feces) is converted into organic fertilizer, CO ₂ and H ₂ O, which are reused for plant culture.	It is used in the later stage of CELSS with high closure to meet the demand for CO ₂ recovery and utilization from solid waste. The configuration scale is determined by the degree of food closure and the amount of biomass solid waste that is to be converted.
11	Brine/feces processing subsystem (brine/feces processing)	Through membrane distillation, the water in urine distillation brine and in feces are further recovered [22].	It is used in conjunction with the urine processing subsystem to recover 98% of the water from urine. As the main means of enhancing the water recovery rate from urine and feces, as well as the removal of inorganic salts such as NaCl from urine, this subsystem will be instrumental in all stages of the lunar base CELSS. Its size is determined by the number of personnel and expands with the increase of the population.
12	Microalgae reactor subsystem (microalgae reactor)	Through microalgae photosynthesis, substances such as CO ₂ and H ₂ O are synthesized into algae bodies, achieving CO ₂ fixation and O ₂ regeneration, and providing algal protein food.	It is used in mid-term lunar base as an alternative to oxygen generation and CO ₂ removal subsystems, to achieve atmospheric regeneration and reserve material resources. In the later stage, it will be used to provide some needed algal proteins. When used as a CO ₂ removal and oxygen regeneration component, the configuration scale is determined by personnel's demand of CO ₂ removal and oxygen. In the case of its utilization as a component for the provision of algal protein, the configuration scale is determined by personnel's demand for algal protein.

food production and waste treatment technologies that have been fully validated in the 4-subject 180-day CELSS integration test [20]. These technologies have undergone the ground simulation environment testing and verification and will be verified on China's space station platform to further enhance technological maturity; (b) The design concept is to gradually increase the system's material closure degree and expand the system scale to meet the life support needs of lunar base with a gradually extended resident time of more than 10 years. The system configuration is composed of physiochemical regeneration technologies, which completes the regeneration of oxygen and water, and transitions to bio-regeneration technologies, which completes the regeneration of oxygen, water, and food. This transition improves the system's material closure degree and self-sustaining ability, thereby reducing dependence on ground supply.

Stage I—Prophase lunar base

At this stage, a life support system suitable for medium- to long-term habitation has been established on the lunar surface, equipped with a complete set of physiochemical regeneration life support system to achieve atmospheric regeneration and water resource recovery. At the same time, according to the operating experience of the International Space Station, a small-scale vegetable production component will be introduced for fresh vegetable and fruit production while providing emotional support for residents [21]. In order to recover as much water as possible, a brine processing subsystem will be added to the existing space station's urine processing and water recovery subsystems to further achieve more than 98% water recovery in urine [22]. In addition, it is necessary to stabilize solid wastes such as inedible plant parts and recover the water contained in them.

Stage II—Preliminary lunar base

Based on stage I, for the 2-year to 5-year residency mission, we will expand the scale of vegetable production component, which could provide all required fresh vegetables in situ. The water resource recycling management system centered on biological treatment technology will be established to provide hygienic and cleaning water for residents. We will achieve stable treatment and water recovery from solid waste such as inedible parts of plants, and recover part of the water in feces through the brine processing subsystem. Furthermore, based on the collection and utilization of lunar ice, a microalgae photobiological reaction system will be constructed, which absorbs CO_2 to produce oxygen through the photosynthesis of microalgae with higher photosynthetic efficiency than plants, and complete the fixation of C/H/O elements. The microalgae photobiological reaction system will accumulate a material basis for the subsequent construction of a higher-level life support system [23–25].

Stage III—Medium lunar base

For the 5- to 10-year residency mission, based on the total vegetable supply, more kinds of plants such as staple crops are introduced to further improve the food closure degree and achieve a food closure degree of 30%–50%; at this time, all atmospheric regeneration is completed through plant components such as vegetables and grains.

Stage IV—Upgrade lunar base

For the residency mission of more than 10 years, the scale of plant components should be further expanded, and the food

closure degree should be increased to more than 60%. Furthermore, the transformation and treatment of solid wastes such as inedible parts and solid excrement of humans should be completed through bioreactors, and recycling and utilization of water and most of carbon in organic matter solid waste. The recycling system including air, water, and waste should be established based on biological regeneration.

For the permanent lunar base, based on the previous construction, a large-scale CELSS will be gradually established to achieve more than 95% food closure, and finally complete a robust circular ecosystem consisting of multi-species with “human–plant–animal–microbe”.

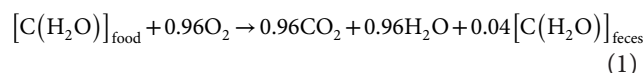
Fig. 1 and Table 2 show the construction process of the lunar base life support system and the functional configuration of each stage.

Design of the overall material flow scheme of CELSS for the lunar base

Basic assumptions, references, and constraints

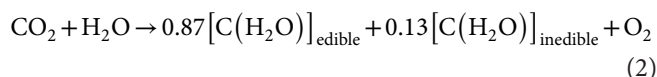
C, H, and O elements are the 3 most important substances in life support systems, with an average proportion of over 94.9% in various organic substances [12,17]. In this paper, we complete the overall design of the material flow of the CELSS for lunar base at different stages and study the quantitative stoichiometric relationships of C, H, and O in the system. We focus on the material flow relationship between bio-regenerative components and physicochemical regenerated components quantitatively. The characteristics of material flow at different stages are quantitatively evaluated. The impact of the introduction of bio-regenerative components on the material flow of the system is also investigated. Therefore, in order to simplify the model, all organic matter (including food, edible parts of plants, inedible parts, solid excrement, organic fertilizer, etc.) are represented by $[\text{C}(\text{H}_2\text{O})]$. The basic assumptions and design constraints are as follows:

According to the average energy intake levels of adult males in the Chinese Residents Dietary Guidelines 2022, as well as the results of 4-subject 180-day integrated test [18,26], the energy input is designed at 2,600 kcal/person/day. The carbohydrate energy is assumed as 3.73 kcal/g, then 697 g of carbohydrates is required. The ratio of feces (dry weight) to the total food intake is assumed as 4%, then the human metabolism model is as follows [Eq. (1)]:



The data related to water demand and drainage of personnel refer to the results of the 4-subject 180-day integrated test and the research report “Life Support Baseline Values and Assumptions Document” by NASA, as shown in Table 3.

Metabolic models for plant components are established based on 2 major categories: vegetables and grains. Lettuce and wheat are used as the model species for vegetables and grains, respectively. The biomass synthesis models are shown in Eqs. (2) and (3), respectively. The design parameters such as edible yield are based on the results of the 4-subject 180-day integrated test and NASA's research report “Life Support Baseline Values and Assumptions Document”, as detailed in Table 4.



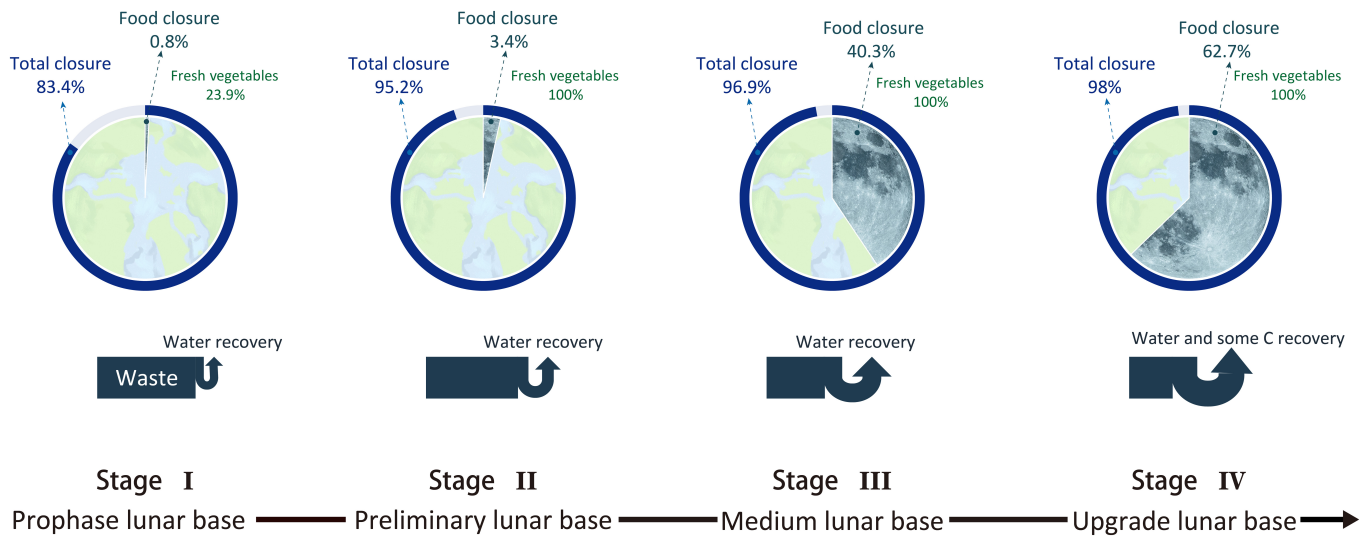
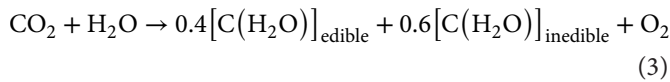
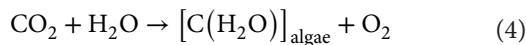


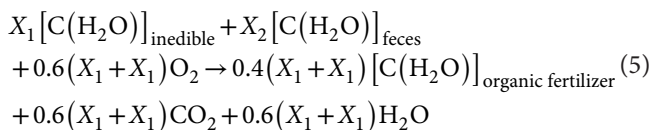
Fig. 1. Overall application plan of CELSS for lunar base.



The biomass synthesis model of microalgae is shown in Eq. (4), and the design parameters such as oxygen and edible yield refer to the results of Poughon et al. [27] and Ai et al. [28]. The biological metabolism of microalgae is calculated at an average of 387 g O₂/m³/day. In addition, the moisture content of the microalgae is set to 95%.



The solid waste biological treatment model is shown in Eq. (5), the carbon conversion rate is calculated as 60%, and the moisture content of initial solid waste is calculated as 55% according to the results of biological treatment of solid waste in the 4-subject 180-day integrated test [29]. Additionally, it is assumed that the organic fertilizer product from the treatment is fully dried and the obtained water enters the water cycle.



The calculation parameters of other subsystems are determined according to the results of previous studies. The urine treatment subsystem recovered 83% of the water in the urine, and the water recovery rate in the urine could reach 98% after further treatment by the urine brine processing subsystem. In the later stage of the lunar base, the feces also enters the brine processing subsystem to recover 90% of the water in the feces [3,22].

When calculating material flow balance, the material inputs and outputs of each functional component or subsystem are assumed to be in a steady state, and thus, the overall material flow of the system is also in a steady state.

The calculation of the coefficient of material closure (C) is determined by the relationship between the rate of material input of the system (m) and the rate of material consumption by crewmembers (M). Alternatively, it is determined by the

total amount of externally replenished matter ($\sum m_{\text{out}}$) and the total amount of material consumed by crewmembers ($\sum M$) over the entire mission time, and the results obtained by the 2 calculation methods are consistent when the system is in a steady state [30].

$$C = (1 - m/M) \times 100\% \quad (6)$$

$$C = \left(1 - \frac{\sum m_{\text{out}}}{\sum M}\right) \times 100\% \quad (7)$$

The above equations apply not only to total material closure, but also to sub-items closure, such as atmosphere, water, and food.

The material flow scheme and quantitative model of the life support system of the lunar base at each stage are described in detail as follows.

Prophase lunar base

The material flow scheme of the prophase lunar base life support system is shown in Fig. 2, which overall inherits the mode of the space station's physicochemical regeneration. Small-scale cultivation of vegetables is introduced to evaluate the impacts of plant components on the physicochemical regeneration subsystem. The regeneration cycle of oxygen and water is completed through these subsystems including urine processing, water recovery, oxygen generation, CO₂ removal, and CO₂ reduction. Additionally, a brine processing subsystem is added to further improve the water recovery rate in urine. Growing about 1 m² of vegetables can provide ~119.6 g of fresh vegetables per person per day, supplementing vitamin and providing ~0.8% of the food (dry weight) supply. The solid waste dewatering subsystem is configured to achieve the stabilization of the inedible parts of plants (vegetable roots, old leaves, etc.), and recover the water from them.

Its technical features include the following:

1. Each person needs 720.1 g of food and 121.1 g of water supply per day. Due to the high cost of earth-to-lunar transportation, the exogenous water introduced into the system through water-containing items such as

Table 2. Functional configuration of CELSS for lunar base at different stages

Stage	Mission duration	General characteristics	Atmospheric regeneration	Water recovery	Food production	Solid waste treatment
Stage I—prophase lunar base	20 weeks to 20 months	•Physicochemical regeneration technology plays a dominant role •Oxygen and water are regenerated •Providing a small number of fresh vegetables	CO ₂ removal, CO ₂ reduction, oxygen generation, vegetable production	Urine processing, water recovery, brine processing	Vegetable production	Waste dewatering
Stage II—preliminary lunar base	2–5 years	•Oxygen is provided by bio-regeneration •Microbial bioreactor participates in the water cycle •Providing all fresh vegetables	CO ₂ removal, microalgae reactor, vegetable production	Urine processing, water recovery, brine processing, wastewater bioreactor	Vegetable production	Waste dewatering
Stage III—medium lunar base	5–10 years	•Oxygen is provided by bio-regeneration •Bioreactor dominates water circulation •Providing all fresh vegetables and a small amount of grain	Vegetable production, grain production	Urine processing, water recovery, brine processing, wastewater bioreactor	Vegetable production, grain production	Waste dewatering
Stage IV—upgrade lunar base	>10 years	•Oxygen is provided by bio-regeneration •Bioreactor dominates water circulation •Solid waste is treated by biological fermentation •Providing all fresh vegetables and majority of the grain	Vegetable production, grain production	Urine processing, water recovery, brine processing, wastewater bioreactor	Vegetable production, grain production	Waste dewatering/grinder, waste bioreactor

food, wet wipes, and wet towels will be greatly reduced. Moreover, both electrolysis and biological photosynthesis consume water to produce oxygen. Then, it is necessary to recover water from metabolites of human and plant as much as possible to reduce replenishment. This requirement is applicable to all the subsequent CELSS regardless of their scale.

- The addition of 1 m² of vegetables had a little influence on the material flow of the physicochemical regeneration system, the water circulation rate increased by $3,750/3,499.4 = 107.2\%$, and the gas circulation rate decreased by $10.1/1,022.2 = 1.0\%$. Since transpiration accelerates the flow of water, the addition of plant components has a relatively large impact on the water cycle.
- Substances discharged or not participating in the material flow cycle include solids and semi-solids such as feces, urine residues, and inedible plant parts, as well as gaseous substances such as CO₂ and CH₄.

Preliminary lunar base

With the increase of available resources such as space and energy, the preliminary lunar base will expand the scale of plant components based on the prophase lunar base, and provide all

the fresh vegetables needed through in situ production. The preliminary lunar base increases the supply of 10.5 kg of water per person per day for hygiene and cleaning. The wastewater bioreactor subsystem is introduced to meet the demand for larger water circulation and achieve classified water supply for different terminals. In addition, water is further recovered from the feces. The specific material flow scheme includes the following (Fig. 3): (a) increasing the plant cultivation area to 4.18 m², providing all the required fresh vegetables with a fresh weight of 500 g per person per day, and reaching the food closure rate of 3.4% (25/726); (b) The urine brine processing subsystem is used to recover 90% of the water in feces and reduce water replenishment; (c) Two wastewater bioreactor subsystems will be added. Bioreactor 1 will serve as the pretreatment unit for urine wastewater, completing the removal of most organic compounds and nitrogen stabilization treatment in urine. This will improve the service life of subsequent urine processing and brine processing subsystems. Bioreactor 2 functions as a transfer station for the water cycle, completing the pretreatment of condensate water, urine distillation water, plant nutrient solution (as required), microalgae culture solution (as required), recovered water from solid waste, urine brine, and feces. It achieves the removal of organic matter and the stabilization of nitrogen (converted into nitrogen or nitrate) in various

Table 3. Assumption and baseline of the human metabolic model

Parameters	Value	Reference
Average daily energy intake/kcal	2,600	[18,26]
Food (carbohydrate) intake (dry weight)/kg	0.726	Calculated based on energy demand and human metabolic model
O ₂ consumption/kg	0.743	Calculated based on food metabolism
CO ₂ emission/kg	1.022	Calculated based on food metabolism
Feces (dry weight)/kg	0.029	Calculated based on food metabolism
Potable water requirements (including food processing water and water in food)/kg	3.313	[44]
Hygiene and cleaning water/kg ^a	10.500	[44,45]
Perspiration/kg	2.244	Calculated based on water balance
Urine/kg	1.400	[19]
Urine flushing water/kg	0.300	[45]
Urine wastewater/kg	1.700	Sum of urine and flushing water
Water in feces/kg	0.087	The water content in feces is calculated as 0.75 [29]
Gray wastewater/kg	10.500	Equal to hygiene and cleaning water consumption
Water production in food metabolizes/kg	0.418	Calculated based on food metabolism (consider when calculating water balance)

^aSupply during preliminary and higher-level lunar bases.

Table 4. Assumption and baseline of plant metabolic model

Crop categories	Edible yield/ (gdw·m ⁻² ·day ⁻¹)	Water content in edible/%	Water content in inedible/%	Transpiration rate/ (kg·m ⁻² ·day ⁻¹)	Harvest index
Vegetables	5.98 ^a	95 ^a	90 ^b	3.75 ^c	0.87 ^a
Grain crops	20 ^b	12 ^b	90 ^b	5.5 ^c	0.4 ^b

^aRefer to Ref. [35].

^bRefer to Ref. [19].

^cRefer to Ref. [44].

wastewaters, producing reclaimed water for plant and microalgae culture. The drinking water and clean water required by humans is further purified by a water recovery subsystem; (d) Adding microalgae reactor to replace electrolysis oxygen generation and CO₂ reduction subsystems, synchronously achieving CO₂ fixation and O₂ regeneration; (e) The solid waste dewatering unit completes the drying and stabilization of inedible plant parts and microalgae bodies simultaneously, and recovers the water from them.

Its technical features include the following:

1. According to the recent exploration results, there might be water ice on the Moon, making it possible to obtain water resources from the lunar surface [31,32]. Therefore, in the case of utilizing in situ water resources on the lunar surface, CO₂ can be captured and fixed through photosynthesis of a microalgae bioreactor, while completing oxygen regeneration to reduce gas emissions.

After dewatering, the harvested algae can be stored as material reserves for building more complex life support and energy generation systems in the future. At the same time, after being transformed by microalgae, lunar ice can become a safe water resource for life support systems. Then, the 2 subsystems of electrolysis oxygen generation and CO₂ removal are switched to a system backup.

2. The volume of water circulation in the system has increased rapidly, to 42.9 l/day, of which the water used by personnel is 13.6 l/day. The advantages of biological treatment technology for the treatment of large volumes and multiple types of wastewaters are becoming increasingly apparent. Moreover, after biological treatment and transformation, the wastewater can meet the plant culture, microalgae culture, urine flushing, and other water needs, to facilitate the realization of the system

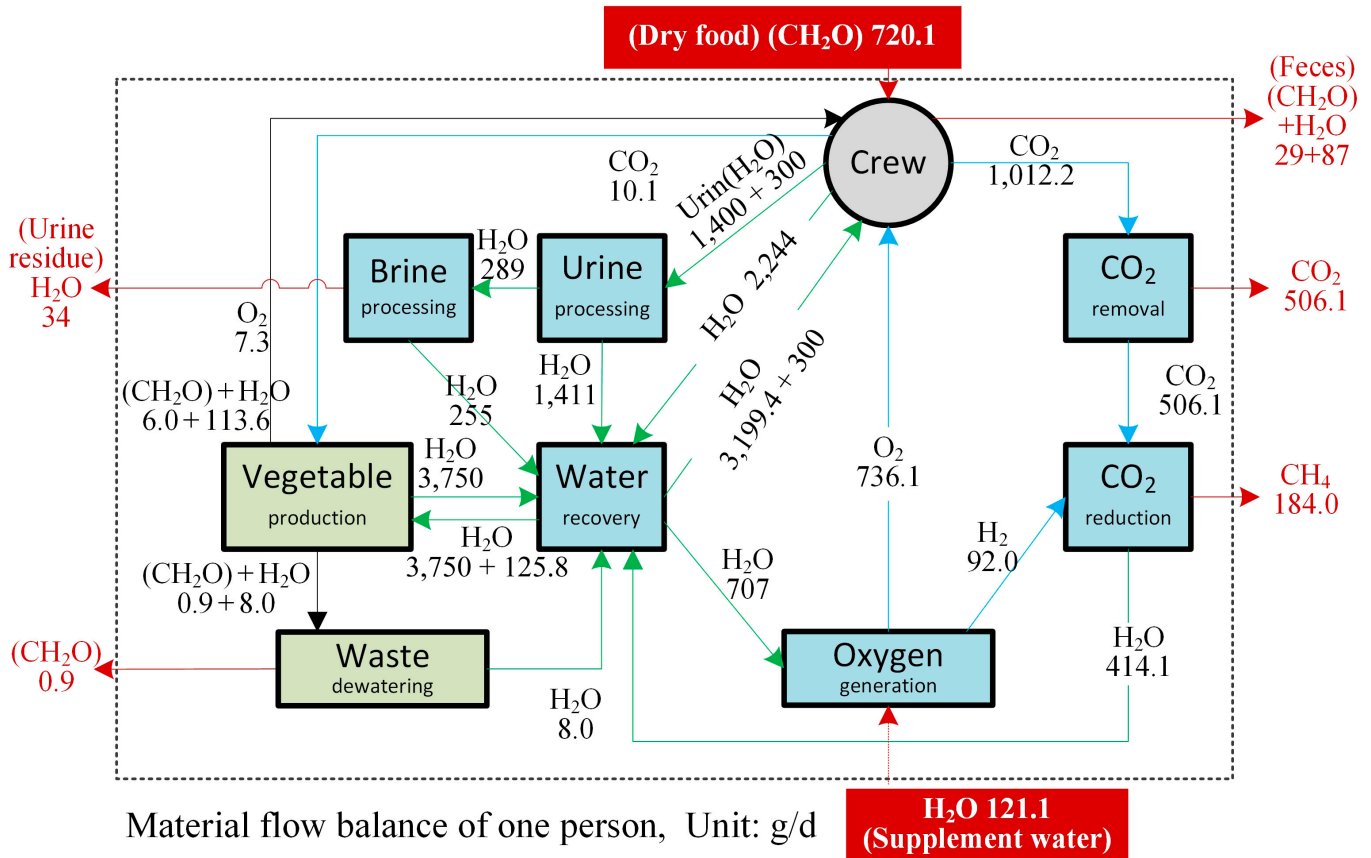


Fig. 2. The material flow scheme of CELSS for the prophase lunar base.

of quality water supply and improve the efficiency of water recycling.

- In addition to the daily requirement of 701 g of dry weight food, the average water supplement per person is 42.8 g/day. The main reason for the decrease in water replenishment is the completion of water recovery in feces. Therefore, it can be considered that the material utilization efficiency of microalgae's oxygen regeneration process is roughly equivalent to the combination of electrolysis oxygen generation and CO₂ reduction. The previously discharged CO₂ and CH₄ substances are converted into algal biomass, which is more convenient for long-term storage and utilization.
- Due to the relatively low gas metabolism ability of plant components at this phase, the CO₂ required for plant growth is completely provided by personnel respiration. Thus, there is no need to oxidize biomass solid waste produced or stored in the system.

Medium lunar base

The material flow scheme for CELSS of the medium lunar base is shown in Fig. 4. To ensure a longer residency and to further reduce the logistics supply, an additional 13.37 m² of grain crops will be added to expand the scale of plant component, resulting in the overall growing area of 17.55 m² per person. During this phase, in addition to completing the production of the required fresh vegetables, a portion of the staple food is also produced and supplied, achieving a food closure rate of

40.3%. All oxygen regeneration is completed through plant photosynthesis, so that the microalgae bioreactor can be used as algal protein food production unit.

Its technical features include the following:

- Plant components dominate oxygen regeneration and provide food.
- The system requires 433.7 g of dry weight food and 42.7 g/day of additional water supply per person per day. The amount of water is roughly equal to the water loss in urine and fecal residue. Improving food closure does not reduce the needs for water replenishment. However, the total water circulation capacity of the system will be greatly increased to 107.3 l per person per day, greatly enhancing the buffering capacity for water shortage.
- If the food closure is further improved, the CO₂ produced by personnel's respiratory metabolism can no longer meet the needs of plant growth. Therefore, it is necessary to activate the solid waste biological treatment devices to accomplish the recovery of carbon and water from solid waste.

Upgrade lunar base

Fig. 5 shows the material flow scheme of CELSS for the upgrade lunar base. The cultivation scale of plant is further increased to improve food closure to 62.7%. The cultivation area of plant components is 25.68 m² per person and includes 4.18 m² of fresh vegetables and 21.50 m² of grain crops. The solid waste biological treatment subsystem will be added to accomplish the

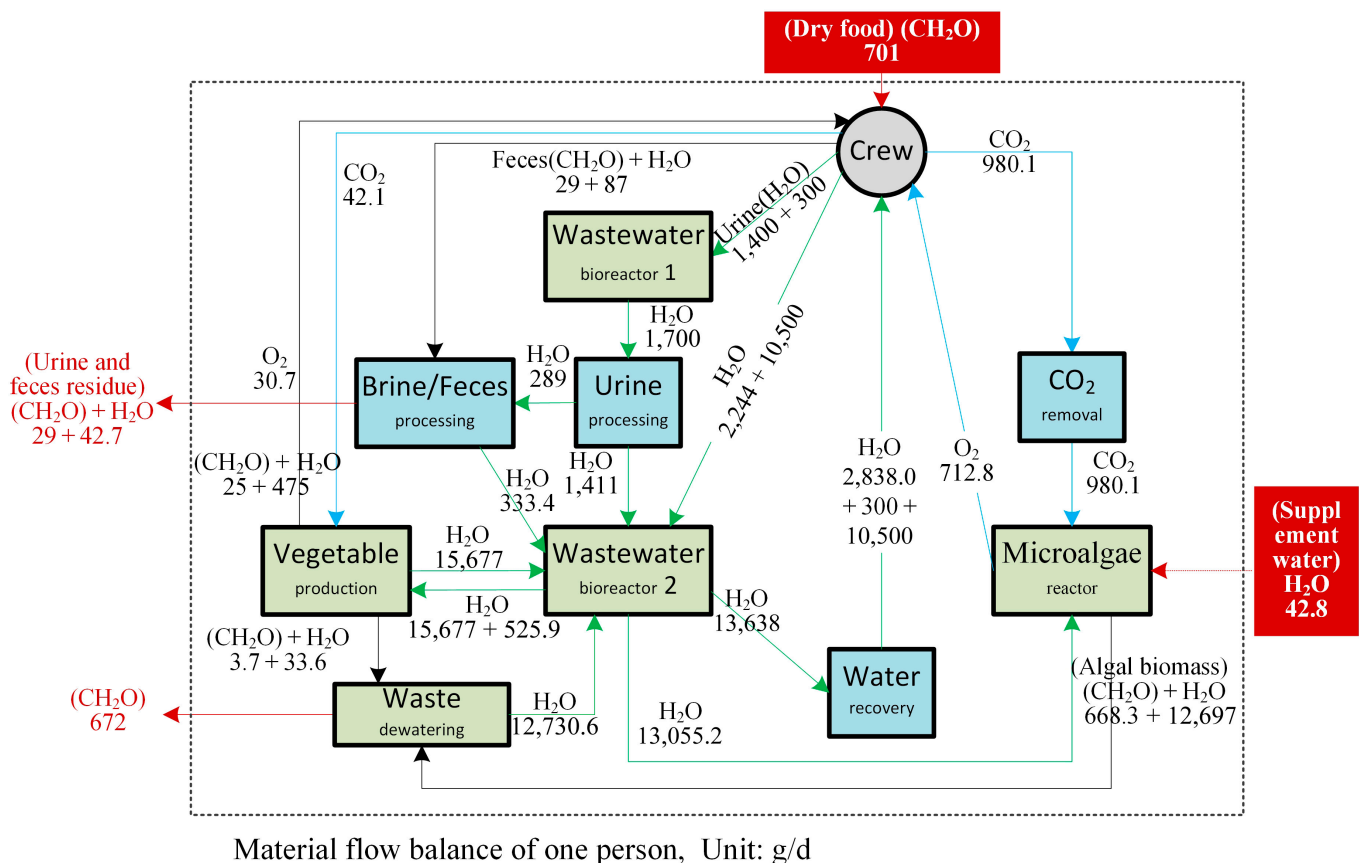


Fig. 3. The material flow scheme of CELSS for the preliminary lunar base.

biotransformation of some inedible plant parts, as well as completing feces collection and biotransformation to replenish CO_2 and water for plant components.

Its technical features include the following:

1. The stability of stool is improved after undergoing solid waste biological treatment. This facilitates the recovery of all the water contained within it, resulting in a further reduction of the system's water replenishment to 34 g per person per day.
2. The waste bioreactor achieves the conversion and recycling of 60% of carbon in the inedible parts of plants. To further improve food closure, it is necessary to add advanced oxidation treatment subsystems such as incineration and wet oxidation to improve the rate and efficiency of carbon conversion in the solid waste.
3. At this phase, the rate of atmospheric regeneration in the system is approximately 1.59 times that of human metabolism. The water cycle is about 11.4 times that of human water consumption, and the total material closure degree of the life support system (based on the total demand of dry weight food, oxygen, and water) reaches 98%.

For a higher-level lunar base, a CELSS of multiple species including large-scale, diverse plants, microalgae, insects, and aquatic animals can be developed to provide complete nutritious food including carbohydrates, plant and animal proteins, fats, vitamins, etc. The CELSS accomplishes total recycling of waste through a combination of biological and physical means

to establish a permanent lunar base with strong buffering capacity and 100% material closure.

Discussion

1. Water recovery is the key point to improving the material closure of the system: From the perspective of existing technical means, both physical and biological regeneration technologies consume water during the regeneration of oxygen. Water can be obtained from inedible plant parts or feces through non-oxidizing technical means such as anaerobic biological fermentation or pyrolysis [33,34]. Water can also be extracted from extraterrestrial in situ resources such as lunar ice. Alternatively, oxygen can be directly collected and extracted from lunar soil to supplement the oxygen demand in the system.
2. The application sequence of plant components: The human body requires a substantial amount of carbohydrates from grain crops. However, the food production process is energy intensive and generates inedible parts such as straw. To complete grain plant and supply staple food, large-scale plant components and corresponding solid waste treatment systems is necessary, which requires a large investment [20]. The staple food can be alternatively supplemented with dry matter, which has a relatively high supply efficiency. Comparatively, fresh vegetables have a short growth cycle and lower energy consumption than grain crops [35]. Fresh vegetables

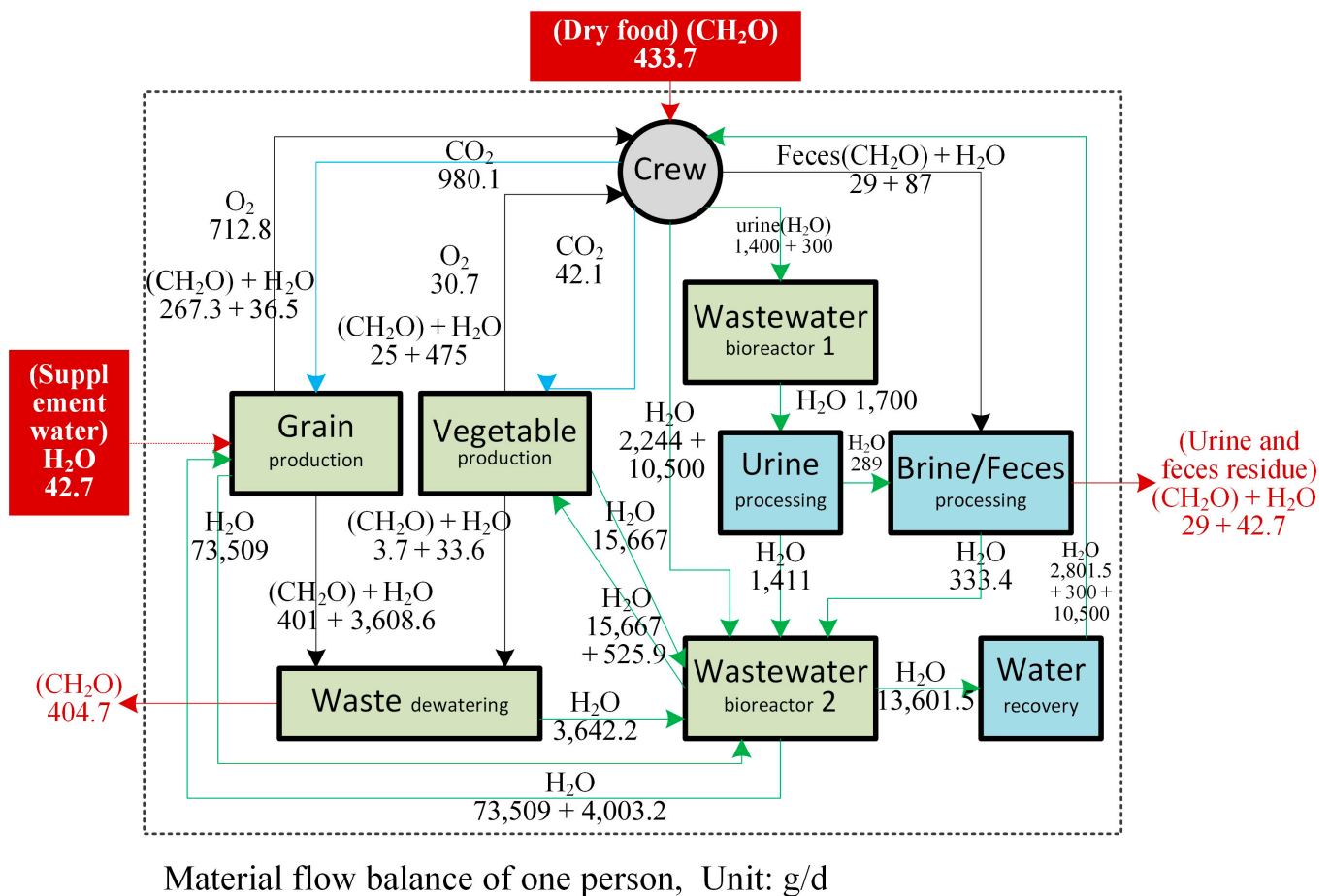


Fig. 4. The material flow scheme of CELSS for the medium lunar base.

not only provide vitamins and dietary fiber necessary for human growth, but also regulate mental emotions through the fresh green environment, as well as visual and gustatory sensations, with relatively high overall efficacy. It is noted that none of the above functions of vegetables can be replaced by dry matter. Therefore, as a plant component in a CELSS, the application of vegetables will take precedence over grain crops.

3. The function and role of microalgae components in the CELSS: The photosynthetic efficiency of microalgae is higher than that of higher plants, and its environment adaptability is stronger. Moreover, the microalgae is an intermediate for substance transformation, which can be used as a food source for humans, aquatic animals, insects, and other animals. It can also be converted into life support substances such as CO₂, H₂O, and H₂ [36,37]. Thus, microalgae have a great application prospect in the CELSS, mainly including the following: (a) As a functional component of the preliminary lunar base, it replaces the CO₂ reduction and electrolysis oxygen production subsystems to complete oxygen regeneration and fix gaseous waste such as CO₂ and CH₄. Microalgae not only provide substance reserves for the establishment of a higher closed CELSS in the future, but also reduce the impact of discharge from early life support system on the original extraterrestrial environment;

(b) Microalgae can supplement and complement higher plants with high-quality algal protein ingredients; (c) As a functional component for the utilization of in situ resources such as lunar ice, microalgae can complete oxygen regeneration through synthesizing water from lunar ice and CO₂ from the system into algal bodies; (d) As a pioneer species of lunar soil amelioration, microalgae can combine microorganisms and higher plants to reform lunar soil and Martian soil, releasing and utilizing the mineral nutrients in them to address the needs for mineral nutrients in plant culture in the extraterrestrial CELSS.

4. Introducing new technologies such as artificial synthesis to improve food production efficiency: With global warming, the concept of 3G biorefinery technology to achieve carbon neutrality was proposed in 2019 and has become a research hotspot [38]. Soon after, a combined chemo-biochemical pathway for the total artificial synthesis of starch from CO₂ and H₂ has been demonstrated in a test tube study, and its energy efficiency and synthesis rate were much higher than those of higher plants [39]. Zheng et al. [40] successfully synthesized glucose and fatty acids from CO₂ by electrocatalysis combined with engineered yeast fermentation. Although there is still a long way to go before engineering applications, the emergence of such technologies is expected to provide

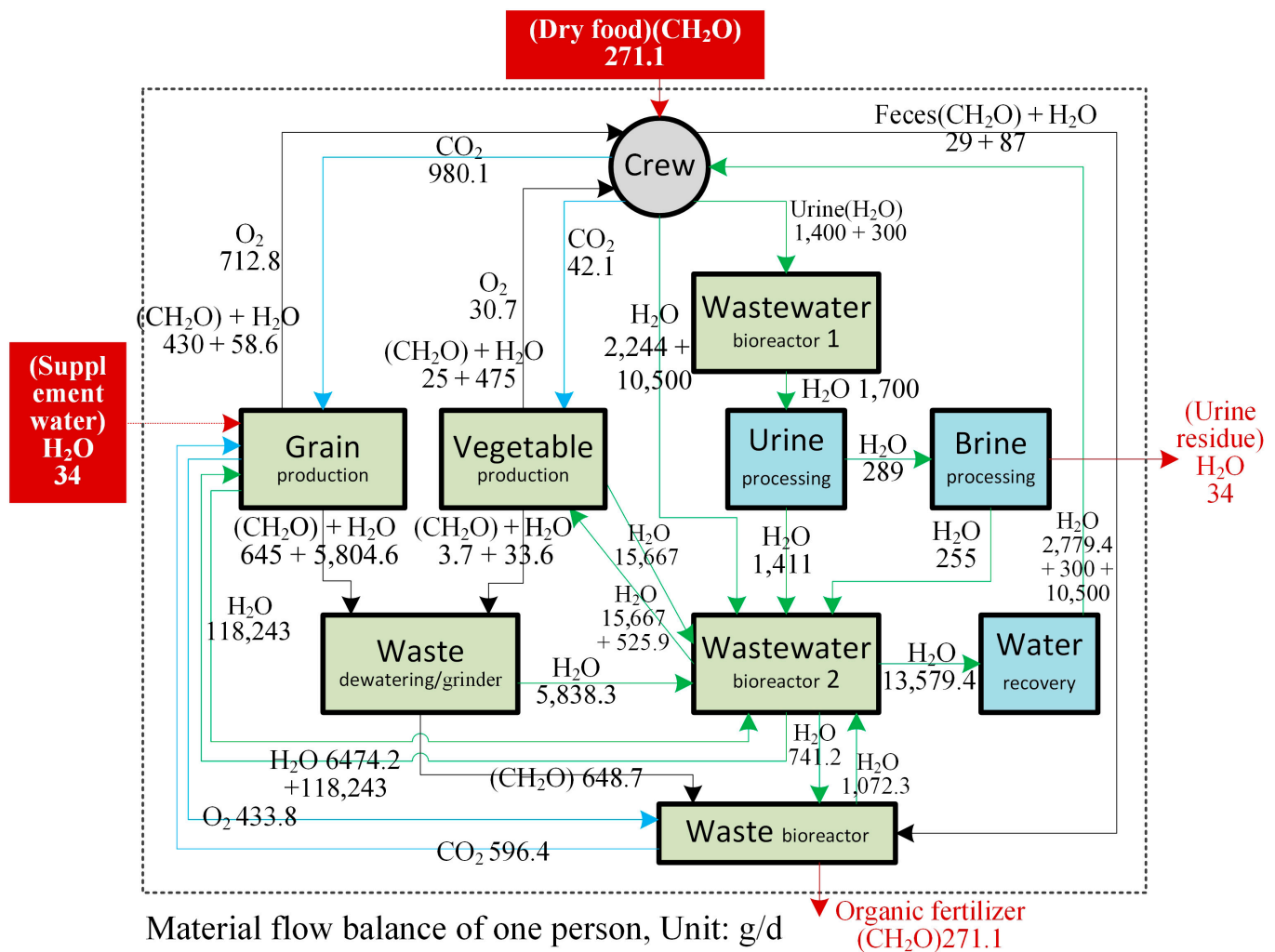


Fig. 5. The material flow scheme of CELSS for the upgrade lunar base.

more efficient options for food production in advanced life support systems in the future.

- Using the space station platform to carry out technical verification and improve technology maturity: Although many experimental studies related to the technology of CELSS have been carried out on the ground [41,42], the technical verification needs to be carried out based on the space station platform in the future. Firstly, technical verification of space plant cultivation must be carried out to solve the problems of production, harvesting, and consumption of fresh vegetables and fruits in orbit. Moreover, core technologies including microalgae cultivation, wastewater biological treatment, and solid waste biological treatment needs to be verified in orbit to address the key issues of stable operation of biological components under practical application environmental conditions. Then, the integrated technical verification of "plant-microalgae-microorganism" small-scale CELSS will be carried out. The small-scale circulation system of water, gas, and solid waste will be established based on cultivation of vegetables and algae. The aims are to verify the material flow circulation technology of bio-regeneration system under space conditions and to solve

the problem of the matching and integrating of bio-regeneration components with existing physicochemical regeneration system. Then, a basis for applications in the lunar environment can be established by technical verification on space station platforms.

- Adopting modular design while strengthening redundant configuration to improve system adaptability and reliability: Each subsystem is designed and manufactured in a modular manner, using unified, standard, and extensible interfaces (including mechanical, electrical, and information interfaces). It can expand the scale and capabilities of the subsystem itself through multiple module combinations to match different stage requirements, while also conveniently integrating into the material flow and connecting to upstream and downstream subsystems to improve the overall adaptability of the system [14,43]. In addition, the following measures are employed to improve the system reliability. In the early stage of the lunar base, the supply of life support materials is mainly achieved through physiochemical regeneration methods. At this time, non-regenerative life support materials from previous manned moon landing missions, including stored oxygen and water,

can serve as system backups to enhance emergency life support capabilities. The higher-level CELSS after the prophase lunar base mainly achieves material circulation through bio-regeneration technology, and the self-sustainability of the system itself is improved. Currently, the physiochemical regeneration subsystems are converted to backup for the bio-regeneration system. This system configuration method has been verified in the 4-subject 180-day CELSS integration test [18,20]. In addition, redundant backups can be individually configured for certain critical subsystems, such as the oxygen generation and water recovery subsystems, to further increase system reliability.

Conclusion

With the goal of improving the engineering application level of the CELSS, this paper completes the quantitative design of the function configuration and material flow model of 4 different levels of lunar base life support systems, and analyzes the material flow characteristics of each system. Based on the existing physicochemical regeneration life support technology of space stations, the bio-regeneration components of different scales including plants, microalgae, and microorganisms are gradually added to establish a life support material flow control system suitable for lunar base in different stages. The present study can provide reference and inspiration for establishing a CELSS for lunar bases at the minimum cost. In the future, the research and application of the life support system technology for the lunar base need to carry out sufficient technical verification to improve the level of biological regeneration technology in actual space environments such as space stations. Furthermore, it is necessary to keep up with the development of new technologies, such as biosynthesis and in situ resource utilization, and introduce new technologies in time to further reduce the cost of system construction and operation.

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Author contributions: Y.D. is the corresponding author, who developed the paper outline and proposed the functional composition and phased implementation of the CELSS. L.Z. is the co-corresponding author, who completed the design of the system material flow model and wrote the first draft of the paper. Q.B. was responsible for the integration scheme for physicochemical regeneration and bio-regeneration components in each stage. W.A. participated in the design and calculation of waste recycling components. Y.T. participated in the design and calculation of quantitative models for vegetable and grain crops, while R.M. participated in the design and calculation of quantitative models for microalgae components. H.F. completed the proofreading and revision of the paper.

Competing interests: The authors declare that they have no competing interests.

Data Availability

The data and materials used to support the findings of this study are available from the corresponding authors upon reasonable request.

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Overall Material Flow Scheme Design of Controlled Ecological Life Support System for Lunar Base

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Recently, the Moon has once again become the focus of the aerospace field. The lunar base is likely to be the next long-term extraterrestrial base for humans after the near-Earth orbit space station. However, the cost of lunar replenishment will substantially increase, and the time required for replenishment will become longer. It is necessary to establish a living system with higher degree of material closure and stronger self-sustainability to provide life support service for residents. The controlled ecological life support system (CELSS) is a feasible way to achieve the circular supply of oxygen, water, and food through biological regeneration. However, there is a practical issue of how to gradually establish a CELSS on the Moon and how to match the bio-regenerative system with the physicochemical system established in the early stage, to reduce the cost of the construction of the lunar life support system. In this paper, based on fully inheriting the physicochemical regeneration technologies of the present near-Earth orbit space station, biological components such as plants, microalgae, and microorganisms were gradually introduced in 4 stages to establish an upgraded CELSS with a total material closure degree of 98%. In addition, the material flux models of C, H, and O elements at different stages of the lunar base were designed and calculated. The results of this paper provide a reference for the construction of future lunar base life support systems in terms of improving the engineering feasibility of CELSS.

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